Yuhao Bao

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RESEARCH INTERESTS

Human-centered AI reliability and evaluation: how AI models interpret crisis-related social media, how their outputs compare with human annotations, and when they should not be trusted in high-stakes settings.

EDUCATION

Brigham Young University, Provo, UT — Expected April 2027
B.S. in Computer Science; Minors in Mathematics and Physics
GPA: 3.85/4.00 (Computer Science: 3.90/4.00)
Relevant coursework: Advanced Deep Learning; Multi-Agent Systems; Database Modeling

PUBLICATIONS AND MANUSCRIPTS

Bao, Y., & Hughes, A. (2026). When Social Media Images Need Words: Measuring Context Gap and Fusion Tax in Crisis Image Captioning. *Proceedings of the International ISCRAM Conference, 23*. Work-in-Progress paper. doi:10.59297/4bar0c32

Bao, Y. What Agreement Hides: LLM Labeling of Hurricane Dorian Tweets (working title). In preparation for the International ISCRAM Conference 2027.

RESEARCH EXPERIENCE

Research Assistant, HCI Research Group, BYU (Advisor: Prof. Amanda Hughes) — Nov 2023 – Present
LLM labeling of Hurricane Dorian tweets (first-author manuscript, in preparation)

- Compared five model versions from three families (Qwen, Gemini, GPT) with historical TREC-IS labels on 355 Hurricane Dorian tweets with 433 photos.
- Similar overall agreement hid systematic differences: Priority agreement ranged from 39.2% to 50.1%, and all five rated the three Critical-labeled tweets as Low or Medium.

Crisis image captioning with vision-language models (ISCRAM 2026)

- Compared Gemini 2.0 Flash, Qwen2.5-VL, and BLIP on 204 crisis image-post pairs with and without post text; adding text raised Gemini's accuracy from 75.5% to 91.7% but also introduced text-driven errors.
- Designed the scoring rubric and error taxonomy and carried out the human coding (Krippendorff's α = 0.69).

Text-to-SQL with Retrieval and LoRA, BYU (Advisor: Prof. Porter Jenkins) — Fall 2025 – Winter 2026

- Built a retrieval-augmented Text-to-SQL pipeline on Qwen3.5-9B with BM25 retrieval and LoRA fine-tuning.
- Implemented Shadow-LoRA (Base-to-Instruct adapter transfer) against a matched Direct-LoRA baseline; exact match rose from 19.47% to 30.49%.

OPEN-SOURCE PROJECTS

modelagree (GitHub): tool for measuring agreement between LLM labels; under development.

crisis-tweet-classifier (GitHub): compares TF-IDF, fine-tuned DistilBERT, and a zero-shot local LLM on HumAID crisis tweets for accuracy, calibration, and cost.

INDUSTRY EXPERIENCE

Software Engineering Intern, Jingus (J.ing), Guangzhou, China — May 2023 – Aug 2023

- Built Shopify storefront components and a PayPal payment integration (REST API, webhooks).

SKILLS

Research methods: rubric and error-taxonomy design, qualitative coding, inter-rater reliability, LLM evaluation
Programming: Python, SQL, Java, C++, JavaScript/TypeScript, Git, Linux
ML and data: PyTorch, Hugging Face Transformers, LoRA/PEFT, BM25 retrieval, pandas, NumPy, PostgreSQL

HONORS AND AFFILIATIONS

BYU travel funding, 2024 SHPE National Convention (selected); member of SHPE, ACM, BYU AI Club